

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester III Examination, 2016-17

Botany

Paper: BOB-301 — Plant Ecology & Physiology

Time: Three hours

Max. Marks: 70

Note: Answer five questions in all, selecting two each from Sections A and B. Question No. 1 is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer/Define the following in not more than 50 words each: [1 × 10 = 10]

- (a) Synecology
- (b) Net primary productivity
- (c) Allogenic succession
- (d) Food web
- (e) Ecological pyramid
- (f) Guttation
- (g) Phototropism
- (h) What is apical dominance?
- (i) Which products of light reaction of photosynthesis are used in the Calvin cycle?
- (j) H⁺-ATPase

SECTION – A

Question 2. Write critical notes on any three of the following: [5 × 3 = 15]

- (a) Atmosphere
- (b) Soil profile
- (c) Qualitative analytical characters of community
- (d) Biotic components of ecosystem

Question 3. Define population. Discuss in brief the various characteristics of a population of organisms. [15]

Question 4. Write descriptive notes on any two of the following: [7.5 × 2 = 15]

- (a) Ecotypes

- (b) Phosphorus cycle
- (c) Universal model of energy flow

Question 5. Define ecological succession. Discuss in detail the course of succession in a hydrosere. [15]

SECTION – B

Question 6. Give an account of the C_4 and CAM pathways of CO_2 fixation in plants. [15]

Question 7. Write descriptive notes on any two of the following: [7.5 × 2 = 15]

- (a) Phytochromes
- (b) Oxidative phosphorylation
- (c) Photorespiration

Question 8. Write short notes on any three of the following: [5 × 3 = 15]

- (a) Translocation of sugar
- (b) Assimilatory reduction of nitrate
- (c) Ion transport
- (d) ABA-induced stomatal closure

Question 9. What are plant growth hormones? How do they differ from plant growth regulators? Give an account of the physiological role of auxins. [15]

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